Power Conversion System (PCS)
IEC 61850 External Communication Protocol

1725kW | 2.5 MW | 3.5 MW

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1 Overview

This protocol applies to communication between the **MPCS series** PCS from Wattox and monitoring software on the IEC 61850 host controller. This protocol enables real-time reading of PCS operational data, related statuses, and control command settings.

2 Control Parameters (CTRL)

2.1 Remote Regulation Parameters (setGGIO1)

Param.	Name	Data Type	Description	Lower Limit	Upper Limit	Default	Unit
A01	Active power setting	Float32	-	–32768	32767	0	1 kW
A02	Reactive power setting	Float32	-	–32768	32767	0	1 kVar
A03	Off-grid line voltage RMS	Float32		0.0	759.0	690.0	0.1 V
A04	PCS on-grid control mode	Float32	0: P-Q mode 1: Constant DC current mode 2: Constant DC voltage mode	0	2	0	1
A05	Power change rate setting	Float32	-	10	30000	10000	1% Pn/s
A06	Constant power factor setting	Float32	-	–1.000	1.000	1.000	0.001
A07	DC bus voltage setting	Float32	-	0	1500.0	1300.0	0.1 V
A08	Current setting in constant current mode	Float32	Sets the bus current (actual value) for constant-current charge/discharge in P-Q mode.	–32768	32767	0	1 A
A09	PCS control command	Float32	0: Off 1: On 2: Reset 3: Standby	0	3	0	1
A10	PCS operation mode	Float32	0: On-grid 1: Off-grid	0	1	0	1
A11	MMS and GOOSE fast control priority	Float32	0: None 1: GOOSE valid 2: MMS valid	0	2	0	1
A12	Reserved						
A13	Reactive power mode setting	Float32	0: Constant reactive power 1: Constant PF 2: Q-U reactive	0	2	0	
A14	Grid-forming function enable	Float32	0: Disable 1: Enable	0	1	0	1
A15	Primary frequency control enable	Float32	0: Disable 1: Enable	0	1	0	1
A16	Primary frequency control dead zone	Float32		0	0.1	0.03	0.01 Hz
A17	Primary frequency control coefficient	Float32		0	65535	50	1
A18	Primary voltage control enable	Float32	0: Disable 1: Enable	0	1	0	1
A19	Primary voltage control dead zone	Float32		0	1.00	0.0	0.01
A20	Primary voltage control coefficient	Float32		0	100.0	1.0	0.1

A21	Virtual inertia coefficient	Float32		0	65535	2000	1
A22	Virtual damping coefficient	Float32		0	65535	1500	1
A23	Virtual excitation coefficient	Float32		0	65535	1200	1
A24	Reserved	Float32		0	65535	0	1
A25	Full-discharge scheduling function selection	Float32	0: Disable 1: Method 1	0	1	0	1
A26	Full-discharge scheduling time setting	Float32		0	65535	0	1

*: A11 priority setting only applies to PCS control commands (on/off/standby), active power setting, and reactive power setting.

2.2 Remote Control Parameters (ctlGAPC1)

Param.	Name	Data Type	Description	Lower Limit	Upper Limit	Default	Unit
B01	Reserved	BOOL	-	0	1	0	1
B02	PCS on/off	BOOL	False: Off True: On	0	1	0	1
B03	Start command	BOOL	False: Invalid True: Valid	0	1	0	1
B04	Stop command	BOOL	False: Invalid True: Valid	0	1	0	1
B05	Fault reset	BOOL	False: Invalid True: Valid	0	1	0	1
B06	Standby command	BOOL	False: Invalid True: Valid	0	1	0	1

3 Measurement Parameters (MEAS)

3.1 Remote Signaling Parameters (staGGIO1)

Param.	Name	Data Type	Description	Lower Limit	Upper Limit	Default	Unit
C01	AC circuit breaker status	BOOL	False: Closed True: Open	0	1	0	1
C02	DC disconnector status (module 1)	BOOL	False: Closed True: Open	0	1	0	1
C03	DC disconnector status (module 2)	BOOL	False: Closed True: Open	0	1	0	1
C04	AC pre-charge contactor status	BOOL	False: Closed True: Open	0	1	0	1
C05	DC pre-charge contactor status (module 1)	BOOL	False: Closed True: Open	0	1	0	1
C06	DC pre-charge contactor status (module 2)	BOOL	False: Closed True: Open	0	1	0	1
C07	Emergency stop button status	BOOL	False: Normal True: Emergency stop	0	1	0	1
C08	PCS on/off status	BOOL	False: Off True: On	0	1	0	1

C09	Command source	BOOL	False: Local True: Remote	0	1	0	1
C10	On-grid charging operation status	BOOL	False: Non-charging operation True: Charging operation	0	1	0	1
C11	On-grid discharging operation status	BOOL	False: Non-discharging operation True: Discharging operation	0	1	0	1
C12	On-grid total operation status	BOOL	False: Non-operating True: Operating	0	1	0	1
C13	Standby status	BOOL	False: Non-standby True: Standby	0	1	0	1
C14	Stop status	BOOL	False: Non-stopped True: Stopped	0	1	0	1
C15	Fault status	BOOL	False: No fault True: Fault occurred	0	1	0	1
C16	Alarm status	BOOL	False: No alarm True: Alarm occurred	0	1	0	1
C17	Off-grid operation status	BOOL	False: Non-off-grid state True: Off-grid operation	0	1	0	1
C18	Lockout status	BOOL	False: None True: Lockout active	0	1	0	1
C19	Charge lockout status	BOOL	False: None True: Charge lockout	0	1	0	1
C20	Discharge lockout status	BOOL	False: None True: Discharge lockout	0	1	0	1

3.2 Remote Measurement Parameters (measGGIO1)

Param.	Name	Data Type	Description	Lower Limit	Upper Limit	Default	Unit
D01	Grid AB line voltage RMS	Float32	-	0	6553.5	0	0.1 V
D02	Grid BC line voltage RMS	Float32	-	0	6553.5	0	0.1 V
D03	Grid CA line voltage RMS	Float32	-	0	6553.5	0	0.1 V
D04	Grid A-phase current RMS	Float32	-	0	65535	0	1 A
D05	Grid B-phase current RMS	Float32	-	0	65535	0	1 A
D06	Grid C-phase current RMS	Float32	-	0	65535	0	1 A
D07	Total active output power	Float32	-	–32768	32767	0	1 kW
D08	Total reactive output power	Float32	-	–32768	32767	0	1 kVar
D09	Output power factor	Float32	-	–1.000	1.000	0	0.001
D10	Battery power	Float32	-	–32768	32767	0	1 kW
D11	Grid apparent power	Float32	-	–32768	32767	0	1 kVA
D12	Current maximum allowable discharging power	Float32	-	0	65535	0	1 kW
D13	Current maximum allowable charging power	Float32	-	0	65535	0	1 kW
D14	Maximum charging power of battery pack	Float32		0	65535	0	1 kW
D15	Maximum discharging power of battery pack	Float32	-	0	65535	0	1 kW
D16	DC side charging current	Float32	-	0	65535	0	1 A
D17	DC side discharging current	Float32	-	0	65535	0	1 A
D18	DC side charging power	Float32		0	65535	0	1 kW
D19	DC side discharging power	Float32	-	0	65535	0	1 kW
D20	AC side total charge active power	Float32	-	0	65535	0	1 kW
D21	AC side total discharge active power	Float32	-	0	65535	0	1 kW
D22	AC side total charge reactive power	Float32	-	0	65535	0	1 kW
D23	AC side total discharge reactive power	Float32	-	0	65535	0	1 kW
D24	Daily discharge amount	Float32	Reset at midnight	0	65535	0	1 kWh
D25	Daily charge amount	Float32	Reset at midnight	0	65535	0	1 kWh
D27	All discharge energy	Float32		0	-	0	1 kWh
D29	All charge energy	Float32	-	0	-	0	1 kWh

D30	PCS system status–bit mode	Int32	Bit 0: Stop Bit 1: Standby Bit 2: Grid-connected charging operation Bit 3: Grid-connected discharging operation Bit 4: Grid-connected zero power operation Bit 5: Off-grid operation Bit 6: Fault status Bit 7: Alarm status	0	65535	0	1
D31	A-phase NTC temperature (module 1)	Float32	-	–3276.8	3276.7	0	0.1°C
D32	B-phase NTC temperature (module 1)	Float32	-	–3276.8	3276.7	0	0.1°C
D33	C-phase NTC temperature (module 1)	Float32	-	–3276.8	3276.7	0	0.1°C
D34	Reactor temperature (module 1)	Float32		–3276.8	3276.7	0	0.1°C
D35	A-phase NTC temperature (module 2)	Float32	-	–3276.8	3276.7	0	0.1°C
D36	B-phase NTC temperature (module 2)	Float32	-	–3276.8	3276.7	0	0.1°C
D37	C-phase NTC temperature (module 2)	Float32	-	–3276.8	3276.7	0	0.1°C
D38	Reactor temperature (module 2)	Float32		–3276.8	3276.7	0	0.1°C
D39	Ambient temperature	Float32		–3276.8	3276.7	0	0.1°C
D40	System insulation impedance	Float32		0	65535	0	1 kΩ
D41	Battery voltage	Float32		–3276.8	3276.7	0	0.1 V
D42	DC current	Float32		–32768	32767	0	1 A
D43	Upper half bus voltage	Float32		–3276.8	3276.7	0	0.1 V
D44	Lower half bus voltage	Float32		–3276.8	3276.7	0	0.1 V
D45	Grid frequency	Float32		0	655.35	0	0.01 Hz
D46	Rated power	Float32		0	65535	1725	1
D47	Reactive power upper limit	Float32		0	32767	0	1
D48	Reactive power lower limit	Float32		–32768	0	0	1
D49	Derating flag	Int32	Bit 0: Derating due to ambient temperature Bit 1: Derating due to coolant inlet temperature Bit 2: Derating due to fan Bit 3: Derating due to reactor temperature Bit 4: Master and slave derating together	0	65535	0	1

D50	DI status (module 1)	Int32	Bit 0: Door status (0: Closed; 1: Open) Bit 1: Insulation detection contactor status (0: Closed; 1: Open) Bit 2: Water level detection status (0: Normal; 1: Fault) Bit 3: DC surge protection device status (0: Normal; 1: Fault) Bit 4: AC surge protection device status (0: Normal; 1: Fault) Bit 5: Water pump power contactor status (0: Closed; 1: Open) Bit 6: Fan power contactor status (0: Closed; 1: Open) Bit 7: Heater contactor status (0: Closed; 1: Open) Bit 8: Reactor temperature control switch status (0: Normal temperature; 1: Over-temperature) Bit 9: Reserved for DI9 Bit 10: Reserved for DI11 Bit 11: Reserved for DI12 Bit 12: Reserved for DI13 Bit 13: Reserved for DI14 Bit 14: Reserved for DI15 Bit 15: Reserved for DI16	0	65535	0	1
D51	DI status (module 2)	Int32	Same as above	0	65535	0	1
D52	Grid-forming PCS internal frequency (reserved)	Float32	-	40.00	55.00	50.00	0.01 Hz
D53	Grid-forming PCS virtual potential (reserved)	Float32	-	0.0	750.0	565.0	0.1 V
D54	PCS system status–value mode	Int32	0: Stopped 1: Standby 2: On-grid charging operation 3: On-grid discharging operation 4: On-grid zero power operation 5: Off-grid operation 6: Fault status	0	65535	0	1
D55	Internal cavity temperature	Float32		–3276.8	3276.7	0	0.1°C
D56	Grid charge A-phase current RMS	Float32	-	0	65535	0	1 A
D57	Grid charge B-phase current RMS	Float32	-	0	65535	0	1 A
D58	Grid charge C-phase current RMS	Float32	-	0	65535	0	1 A
D59	Grid discharge A-phase current RMS	Float32	-	0	65535	0	1 A
D60	Grid discharge B-phase current RMS	Float32	-	0	65535	0	1 A
D61	Grid discharge C-phase current RMS	Float32	-	0	65535	0	1 A

3.3 BMS Parameters (measGGIO2)

Param.	Name	Data Type	Description	Lower Limit	Upper Limit	Default	Unit
E01	Battery pack SOC	Float32	-	0	100.0	-	0.1%
E02	Maximum charging power of battery pack	Float32	-	0	6553.5	-	0.1 kW
E03	Maximum discharging power of battery pack	Float32	-	0	6553.5	-	0.1 kW
E04	BMS operation status	Int32	00: Initial 01: Normal 02: Charge prohibited 03: Discharge prohibited 04: Alarm 05: Standby 06: Fault 07: Reserved	0	65535	-	1
E05	BMS status word	Int32	See Table 1.	0	65535	-	1
E06	Total voltage of battery pack	Float32	-	0	6553.5	-	0.1 V
E07	Total current of battery pack	Float32	The value is negative during charging and positive during discharging.	–3276.8	3276.7	-	0.1 A
E08	Maximum charging current of battery pack	Float32	-	0	6553.5	-	0.1 A
E09	Maximum discharging current of battery pack	Float32	-	0	6553.5	-	0.1 A
E10	Highest cell voltage	Float32	-	0	65.535	-	0.001 V
E11	Lowest cell voltage	Float32	-	0	65.535	-	0.001 V
E12	Battery pack SOH	Float32	-	0	100.0	-	0.1%
E13	Reserved						
E14	Reserved						

Table 1 BMS status word

Name	Bit	Parameter	Description
BMS status word	0	BMS no communication	0: Invalid; 1: Occurred
	1	BMS fault	0: Invalid; 1: Occurred
	2	SOC above the upper limit	0: Invalid; 1: Occurred
	3	SOC below the lower limit	0: Invalid; 1: Occurred
	4	BMS charge prohibited	0: Invalid; 1: Occurred
	5	BMS discharge prohibited	0: Invalid; 1: Occurred
	6	BMS initialization	0: Invalid; 1: Occurred
	7	BMS alarm	0: Invalid; 1: Occurred
	8	BMS charge and discharge prohibited	0: Invalid; 1: Occurred
	9	Reserved	-
	10	Reserved	-
	11	Reserved	-
	12	Reserved	-
	13	Reserved	-
	14	Reserved	-
	15	Reserved	-

[1] When the BMS status word value is 0 (that is, all bits are 0), it indicates that the **BMS system is normal**.

3.4 Fault Status 1 (alarmGGIO1)

Param.	Fault Name	Data Type	Description
F01	Storage battery polarity connected reversely	BOOL	0: No fault; 1: Fault occurred
F02	Storage battery voltage too high	BOOL	0: No fault; 1: Fault occurred
F03	Storage battery voltage too low	BOOL	0: No fault; 1: Fault occurred
F04	Battery voltage too high	BOOL	0: No fault; 1: Fault occurred
F05	Battery voltage too low	BOOL	0: No fault; 1: Fault occurred
F06	BMS system fault	BOOL	0: No fault; 1: Fault occurred
F07	BMS pre-charge fault	BOOL	0: No fault; 1: Fault occurred
F08	BMS main contactor fault	BOOL	0: No fault; 1: Fault occurred
F09	BMS insulation fault	BOOL	0: No fault; 1: Fault occurred
F10	DC side half-bus hardware overvoltage	BOOL	0: No fault; 1: Fault occurred
F11	DC side full-bus software overvoltage	BOOL	0: No fault; 1: Fault occurred
F12	DC side full-bus software undervoltage	BOOL	0: No fault; 1: Fault occurred

F13	DC side half-bus voltage variation too large	BOOL	0: No fault; 1: Fault occurred
F14	AC side hardware overcurrent	BOOL	0: No fault; 1: Fault occurred
F15	AC side software overcurrent	BOOL	0: No fault; 1: Fault occurred
F16	DC side hardware overcurrent	BOOL	0: No fault; 1: Fault occurred
F17	DC side software overcurrent	BOOL	0: No fault; 1: Fault occurred
F18	IGBT Desat fault	BOOL	0: No fault; 1: Fault occurred
F19	IGBT block fault	BOOL	0: No fault; 1: Fault occurred
F20	IGBT over-temperature	BOOL	0: No fault; 1: Fault occurred
F21	AC side A-phase Hall disconnected	BOOL	0: No fault; 1: Fault occurred
F22	AC side B-phase Hall disconnected	BOOL	0: No fault; 1: Fault occurred
F23	AC side C-phase Hall disconnected	BOOL	0: No fault; 1: Fault occurred
F24	DC side current Hall disconnected	BOOL	0: No fault; 1: Fault occurred
F25	Detection board model mismatch	BOOL	0: No fault; 1: Fault occurred
F26	DC side pre-charge contactor fault	BOOL	0: No fault; 1: Fault occurred
F27	DC side switch fault	BOOL	0: No fault; 1: Fault occurred
F28	AC side pre-charge contactor fault	BOOL	0: No fault; 1: Fault occurred
F29	AC side switch fault	BOOL	0: No fault; 1: Fault occurred
F30	CBC overcurrent	BOOL	0: No fault; 1: Fault occurred
F31	Device overload	BOOL	0: No fault; 1: Fault occurred
F32	Grid AC overvoltage	BOOL	0: No fault; 1: Fault occurred

3.5 Fault Status 2 (alarmGGIO2)

Param.	Fault Name	Data Type	Description
G01	Grid AC undervoltage	BOOL	0: No fault; 1: Fault occurred
G02	Grid AC overfrequency	BOOL	0: No fault; 1: Fault occurred
G03	Grid AC underfrequency	BOOL	0: No fault; 1: Fault occurred
G04	Output phase loss	BOOL	0: No fault; 1: Fault occurred
G05	Grid voltage sequence incorrect	BOOL	0: No fault; 1: Fault occurred
G06	Phase lock error	BOOL	0: No fault; 1: Fault occurred
G07	Grid voltage imbalance	BOOL	0: No fault; 1: Fault occurred
G08	AC overvoltage in off-grid mode	BOOL	0: No fault; 1: Fault occurred
G09	AC undervoltage in off-grid mode	BOOL	0: No fault; 1: Fault occurred
G10	Islanding fault	BOOL	0: No fault; 1: Fault occurred
G11	Current control deviation too large	BOOL	0: No fault; 1: Fault occurred

G12	Bus soft-start failure	BOOL	0: No fault; 1: Fault occurred
G13	DC component in the current beyond limit on the AC side	BOOL	0: No fault; 1: Fault occurred
G14	AC switch closing voltage mismatch	BOOL	0: No fault; 1: Fault occurred
G15	AD zero drift too large	BOOL	0: No fault; 1: Fault occurred
G16	Low insulation resistance	BOOL	0: No fault; 1: Fault occurred
G17	Insulation detection failed	BOOL	0: No fault; 1: Fault occurred
G18	Internal cavity over-temperature	BOOL	0: No fault; 1: Fault occurred
G19	Internal cavity over-temperature alarm	BOOL	0: No fault; 1: Fault occurred
G20	Internal cavity temperature low	BOOL	0: No fault; 1: Fault occurred
G21	Filter capacitor over-temperature	BOOL	0: No fault; 1: Fault occurred
G22	Filter capacitor over-temperature alarm	BOOL	0: No fault; 1: Fault occurred
G23	Filter capacitor temperature low	BOOL	0: No fault; 1: Fault occurred
G24	Auxiliary transformer over-temperature	BOOL	0: No fault; 1: Fault occurred
G25	Auxiliary transformer over-temperature alarm	BOOL	0: No fault; 1: Fault occurred
G26	Auxiliary transformer temperature low	BOOL	0: No fault; 1: Fault occurred
G27	High ambient temperature	BOOL	0: No fault; 1: Fault occurred
G28	High ambient temperature alarm	BOOL	0: No fault; 1: Fault occurred
G29	Low ambient temperature	BOOL	0: No fault; 1: Fault occurred
G30	IGBT module A-phase over-temperature	BOOL	0: No fault; 1: Fault occurred
G31	IGBT module B-phase over-temperature	BOOL	0: No fault; 1: Fault occurred
G32	IGBT module C-phase over-temperature	BOOL	0: No fault; 1: Fault occurred

3.6 Fault Status 3 (alarmGGIO3)

Param.	Fault Name	Data Type	Description
H01	IGBT temperature sampling error	BOOL	0: No fault; 1: Fault occurred
H02	IGBT over-temperature alarm	BOOL	0: No fault; 1: Fault occurred
H03	Reactor over-temperature	BOOL	0: No fault; 1: Fault occurred
H04	Reactor over-temperature alarm	BOOL	0: No fault; 1: Fault occurred
H05	Reactor temperature low	BOOL	0: No fault; 1: Fault occurred
H06	Surge protection device fault	BOOL	0: No fault; 1: Fault occurred
H07	Parallel communication fault	BOOL	0: No fault; 1: Fault occurred
H08	Abnormal stop of the master	BOOL	0: No fault; 1: Fault occurred
H09	Carrier synchronization failed	BOOL	0: No fault; 1: Fault occurred
H10	Internal cavity fan fault	BOOL	0: No fault; 1: Fault occurred
H11	Circulating fan fault	BOOL	0: No fault; 1: Fault occurred
H12	Reactor fan fault	BOOL	0: No fault; 1: Fault occurred
H13	IGBT fan fault	BOOL	0: No fault; 1: Fault occurred
H14	Reserved	BOOL	0: No fault; 1: Fault occurred
H15	CAN communication fault between control board and communication board	BOOL	0: No fault; 1: Fault occurred
H16	HMI communication fault	BOOL	0: No fault; 1: Fault occurred
H17	BMS communication fault	BOOL	0: No fault; 1: Fault occurred
H18	EMS connection timeout 1 (external bus)	BOOL	0: No fault; 1: Fault occurred
H19	EMS connection timeout 1 (internal communication)	BOOL	0: No fault; 1: Fault occurred
H20	EMS connection timeout 2 (external bus)	BOOL	0: No fault; 1: Fault occurred
H21	EMS connection timeout 2 (internal communication)	BOOL	0: No fault; 1: Fault occurred
H22	EMS communication fault (dual network)	BOOL	0: No fault; 1: Fault occurred
H23	EEPROM fault	BOOL	0: No fault; 1: Fault occurred
H24	EEPROM alarm	BOOL	0: No fault; 1: Fault occurred
H25	Black box error	BOOL	0: No fault; 1: Fault occurred
H26	FPGA data exchange error	BOOL	0: No fault; 1: Fault occurred
H27	RTC error	BOOL	0: No fault; 1: Fault occurred
H28	220 V control power disconnected	BOOL	0: No fault; 1: Fault occurred
H29	Leakage current Hall self-test failed	BOOL	0: No fault; 1: Fault occurred
H30	Residual abrupt leakage current overlimit	BOOL	0: No fault; 1: Fault occurred
H31	Continuous residual leakage current overlimit	BOOL	0: No fault; 1: Fault occurred
H32	Local emergency stop fault	BOOL	0: No fault; 1: Fault occurred

3.7 Fault Status 4 (alarmGGIO4)

Param.	Fault Name	Data Type	Description
I01	Remote emergency stop fault	BOOL	0: No fault; 1: Fault occurred
I02	BMS emergency stop fault	BOOL	0: No fault; 1: Fault occurred
I03	Phase-to-PE fault	BOOL	0: No fault; 1: Fault occurred
I04	Capacitor voltage resonance fault	BOOL	0: No fault; 1: Fault occurred
I05	High humidity	BOOL	0: No fault; 1: Fault occurred
I06	Humidity detection disconnected	BOOL	0: No fault; 1: Fault occurred
I07	Temperature too high	BOOL	0: No fault; 1: Fault occurred
I08	Temperature detection disconnected	BOOL	0: No fault; 1: Fault occurred
I09	Fuse fault	BOOL	0: No fault; 1: Fault occurred
I10	Cabinet door fault	BOOL	0: No fault; 1: Fault occurred
I11	Water pump locked-rotor	BOOL	0: No fault; 1: Fault occurred
I12	Water pump speed regulation signal absent	BOOL	0: No fault; 1: Fault occurred
I13	Water pump idling	BOOL	0: No fault; 1: Fault occurred
I14	Water cooling system hydraulic fault	BOOL	0: No fault; 1: Fault occurred
I15	Fan power sequence incorrect	BOOL	0: No fault; 1: Fault occurred
I16	Fan power disconnected	BOOL	0: No fault; 1: Fault occurred
I17	Three-phase fan fault	BOOL	0: No fault; 1: Fault occurred
I18	Heater contactor fault	BOOL	0: No fault; 1: Fault occurred
I19	Water pump power disconnected	BOOL	0: No fault; 1: Fault occurred
I20	Software version inconsistent between master and slave	BOOL	0: No fault; 1: Fault occurred
I21	Dynamic environment fault	BOOL	0: No fault; 1: Fault occurred
I22	DC side midpoint sampling disconnected	BOOL	0: No fault; 1: Fault occurred
I23	DC side voltage sampling disconnected	BOOL	0: No fault; 1: Fault occurred

I24	AC and DC voltage mismatch	BOOL	0: No fault; 1: Fault occurred
I25	Grid voltage mismatch	BOOL	0: No fault; 1: Fault occurred
I26	Heat exchanger fan fault	BOOL	0: No fault; 1: Fault occurred
I27	DC circuit breaker fan fault	BOOL	0: No fault; 1: Fault occurred
I28	AC circuit breaker fan fault	BOOL	0: No fault; 1: Fault occurred
I29	Master and slave identification fault	BOOL	0: No fault; 1: Fault occurred
I30	Power inconsistent between master and slave	BOOL	0: No fault; 1: Fault occurred
I31	IGBT temperature low	BOOL	0: No fault; 1: Fault occurred
I32	Zero sequence loop current too large	BOOL	0: No fault; 1: Fault occurred

3.8 Fault Status 5 (alarmGGIO5)

Param.	Fault Name	Data Type	Description
J01	FPGA side fault	BOOL	0: No fault; 1: Fault occurred
J02	Fault of parallel devices in cabinet	BOOL	0: No fault; 1: Fault occurred
J03	Fiber communication interrupted	BOOL	0: No fault; 1: Fault occurred
J04	Fiber communication CRC fault	BOOL	0: No fault; 1: Fault occurred
J05	Control mode inconsistent	BOOL	0: No fault; 1: Fault occurred
J06	Capacitor harmonic overlimit	BOOL	0: No fault; 1: Fault occurred
J07	Fan pump and water pump switch error	BOOL	0: No fault; 1: Fault occurred
J08	PCS interaction fault	BOOL	0: No fault; 1: Fault occurred
J09	Capacitor harmonic sampling disconnected	BOOL	0: No fault; 1: Fault occurred
J10	Reserved	BOOL	0: No fault; 1: Fault occurred
J11	Reserved	BOOL	0: No fault; 1: Fault occurred
J12	Reserved	BOOL	0: No fault; 1: Fault occurred
J13	Reserved	BOOL	0: No fault; 1: Fault occurred
J14	Reserved	BOOL	0: No fault; 1: Fault occurred
J15	Reserved	BOOL	0: No fault; 1: Fault occurred
J16	Reserved	BOOL	0: No fault; 1: Fault occurred
J17	Reserved	BOOL	0: No fault; 1: Fault occurred
J18	Reserved	BOOL	0: No fault; 1: Fault occurred
J19	Reserved	BOOL	0: No fault; 1: Fault occurred
J20	Reserved	BOOL	0: No fault; 1: Fault occurred
J21	Reserved	BOOL	0: No fault; 1: Fault occurred
J22	Reserved	BOOL	0: No fault; 1: Fault occurred

J23	Reserved	BOOL	0: No fault; 1: Fault occurred
J24	Reserved	BOOL	0: No fault; 1: Fault occurred
J25	Reserved	BOOL	0: No fault; 1: Fault occurred
J26	Reserved	BOOL	0: No fault; 1: Fault occurred
J27	Reserved	BOOL	0: No fault; 1: Fault occurred
J28	Reserved	BOOL	0: No fault; 1: Fault occurred
J29	Reserved	BOOL	0: No fault; 1: Fault occurred
J30	Reserved	BOOL	0: No fault; 1: Fault occurred
J31	Reserved	BOOL	0: No fault; 1: Fault occurred
J32	Reserved	BOOL	0: No fault; 1: Fault occurred

* Note: Fault statuses are continuously updated; for faults not listed, refer to HMI display or consult PCS manufacturer.

3.9 Alarm Status 1 (warningGGIO1)

Param.	Fault Name	Data Type	Description
K01	Storage battery polarity connected reversely	BOOL	0: No alarm; 1: Alarm occurred
K02	Storage battery voltage too high	BOOL	0: No alarm; 1: Alarm occurred
K03	Storage battery voltage too low	BOOL	0: No alarm; 1: Alarm occurred
K04	Battery voltage too high	BOOL	0: No alarm; 1: Alarm occurred
K05	Battery voltage too low	BOOL	0: No alarm; 1: Alarm occurred
K06	BMS system fault	BOOL	0: No alarm; 1: Alarm occurred
K07	BMS pre-charge fault	BOOL	0: No alarm; 1: Alarm occurred
K08	BMS main contactor fault	BOOL	0: No alarm; 1: Alarm occurred
K09	BMS insulation fault	BOOL	0: No alarm; 1: Alarm occurred
K10	DC side half-bus hardware overvoltage	BOOL	0: No alarm; 1: Alarm occurred
K11	DC side full-bus software overvoltage	BOOL	0: No alarm; 1: Alarm occurred
K12	DC side full-bus software undervoltage	BOOL	0: No alarm; 1: Alarm occurred
K13	DC side half-bus voltage variation too large	BOOL	0: No alarm; 1: Alarm occurred
K14	AC side hardware overcurrent	BOOL	0: No alarm; 1: Alarm occurred
K15	AC side software overcurrent	BOOL	0: No alarm; 1: Alarm occurred
K16	DC side hardware overcurrent	BOOL	0: No alarm; 1: Alarm occurred
K17	DC side software overcurrent	BOOL	0: No alarm; 1: Alarm occurred
K18	IGBT Desat fault	BOOL	0: No alarm; 1: Alarm occurred
K19	IGBT block fault	BOOL	0: No alarm; 1: Alarm occurred

K20	IGBT over-temperature	BOOL	0: No alarm; 1: Alarm occurred
K21	AC side A-phase Hall disconnected	BOOL	0: No alarm; 1: Alarm occurred
K22	AC side B-phase Hall disconnected	BOOL	0: No alarm; 1: Alarm occurred
K23	AC side C-phase Hall disconnected	BOOL	0: No alarm; 1: Alarm occurred
K24	DC side current Hall disconnected	BOOL	0: No alarm; 1: Alarm occurred
K25	Detection board model mismatch	BOOL	0: No alarm; 1: Alarm occurred
K26	DC side pre-charge contactor fault	BOOL	0: No alarm; 1: Alarm occurred
K27	DC side switch fault	BOOL	0: No alarm; 1: Alarm occurred
K28	AC side pre-charge contactor fault	BOOL	0: No alarm; 1: Alarm occurred
K29	AC side switch fault	BOOL	0: No alarm; 1: Alarm occurred
K30	CBC overcurrent	BOOL	0: No alarm; 1: Alarm occurred
K31	Device overload	BOOL	0: No alarm; 1: Alarm occurred
K32	Grid AC overvoltage	BOOL	0: No alarm; 1: Alarm occurred

3.10 Alarm Status 2 (warningGGIO2)

Param.	Fault Name	Data Type	Description
L01	Grid AC undervoltage	BOOL	0: No alarm; 1: Alarm occurred
L02	Grid AC overfrequency	BOOL	0: No alarm; 1: Alarm occurred
L03	Grid AC underfrequency	BOOL	0: No alarm; 1: Alarm occurred
L04	Output phase loss	BOOL	0: No alarm; 1: Alarm occurred
L05	Grid voltage sequence incorrect	BOOL	0: No alarm; 1: Alarm occurred
L06	Phase lock error	BOOL	0: No alarm; 1: Alarm occurred
L07	Grid voltage imbalance	BOOL	0: No alarm; 1: Alarm occurred
L08	AC overvoltage in off-grid mode	BOOL	0: No alarm; 1: Alarm occurred
L09	AC undervoltage in off-grid mode	BOOL	0: No alarm; 1: Alarm occurred
L10	Islanding fault	BOOL	0: No alarm; 1: Alarm occurred
L11	Current control deviation too large	BOOL	0: No alarm; 1: Alarm occurred
L12	Bus soft-start failure	BOOL	0: No alarm; 1: Alarm occurred
L13	DC component in the current beyond limit on the AC side	BOOL	0: No alarm; 1: Alarm occurred
L14	AC switch closing voltage mismatch	BOOL	0: No alarm; 1: Alarm occurred
L15	AD zero drift too large	BOOL	0: No alarm; 1: Alarm occurred
L16	Low insulation resistance	BOOL	0: No alarm; 1: Alarm occurred
L17	Insulation detection failed	BOOL	0: No alarm; 1: Alarm occurred
L18	Internal cavity over-temperature	BOOL	0: No alarm; 1: Alarm occurred

L19	Internal cavity over-temperature alarm	BOOL	0: No alarm; 1: Alarm occurred
L20	Internal cavity temperature low	BOOL	0: No alarm; 1: Alarm occurred
L21	Filter capacitor over-temperature	BOOL	0: No alarm; 1: Alarm occurred
L22	Filter capacitor over-temperature alarm	BOOL	0: No alarm; 1: Alarm occurred
L23	Filter capacitor temperature low	BOOL	0: No alarm; 1: Alarm occurred
L24	Auxiliary transformer over-temperature	BOOL	0: No alarm; 1: Alarm occurred
L25	Auxiliary transformer over-temperature alarm	BOOL	0: No alarm; 1: Alarm occurred
L26	Auxiliary transformer temperature low	BOOL	0: No alarm; 1: Alarm occurred
L27	High ambient temperature	BOOL	0: No alarm; 1: Alarm occurred
L28	High ambient temperature alarm	BOOL	0: No alarm; 1: Alarm occurred
L29	Low ambient temperature	BOOL	0: No alarm; 1: Alarm occurred
L30	IGBT module A-phase over-temperature	BOOL	0: No alarm; 1: Alarm occurred
L31	IGBT module B-phase over-temperature	BOOL	0: No alarm; 1: Alarm occurred
L32	IGBT module C-phase over-temperature	BOOL	0: No alarm; 1: Alarm occurred

3.11 Alarm Status 3 (warningGGIO3)

Param.	Fault Name	Data Type	Description
M01	IGBT temperature sampling error	BOOL	0: No alarm; 1: Alarm occurred
M02	IGBT over-temperature alarm	BOOL	0: No alarm; 1: Alarm occurred
M03	Reactor over-temperature	BOOL	0: No alarm; 1: Alarm occurred
M04	Reactor over-temperature alarm	BOOL	0: No alarm; 1: Alarm occurred
M05	Reactor temperature low	BOOL	0: No alarm; 1: Alarm occurred
M06	Surge protection device fault	BOOL	0: No alarm; 1: Alarm occurred
M07	Parallel communication fault	BOOL	0: No alarm; 1: Alarm occurred
M08	Abnormal stop of the master	BOOL	0: No alarm; 1: Alarm occurred
M09	Carrier synchronization failed	BOOL	0: No alarm; 1: Alarm occurred
M10	Internal cavity fan fault	BOOL	0: No alarm; 1: Alarm occurred
M11	Circulating fan fault	BOOL	0: No alarm; 1: Alarm occurred
M12	Reactor fan fault	BOOL	0: No alarm; 1: Alarm occurred
M13	IGBT fan fault	BOOL	0: No alarm; 1: Alarm occurred
M14	Reserved	BOOL	0: No alarm; 1: Alarm occurred
M15	CAN communication fault between control board and communication board	BOOL	0: No alarm; 1: Alarm occurred
M16	HMI communication fault	BOOL	0: No alarm; 1: Alarm occurred
M17	BMS communication fault	BOOL	0: No alarm; 1: Alarm occurred

M18	Reserved	BOOL	0: No alarm; 1: Alarm occurred
M19	Reserved	BOOL	0: No alarm; 1: Alarm occurred
M20	Reserved	BOOL	0: No alarm; 1: Alarm occurred
M21	Reserved	BOOL	0: No alarm; 1: Alarm occurred
M22	EMS communication fault (dual network)	BOOL	0: No alarm; 1: Alarm occurred
M23	EEPROM fault	BOOL	0: No alarm; 1: Alarm occurred
M24	EEPROM alarm	BOOL	0: No alarm; 1: Alarm occurred
M25	Black box error	BOOL	0: No alarm; 1: Alarm occurred
M26	FPGA data exchange error	BOOL	0: No alarm; 1: Alarm occurred
M27	RTC error	BOOL	0: No alarm; 1: Alarm occurred
M28	220 V control power disconnected	BOOL	0: No alarm; 1: Alarm occurred
M29	Leakage current Hall self-test failed	BOOL	0: No alarm; 1: Alarm occurred
M30	Residual abrupt leakage current overlimit	BOOL	0: No alarm; 1: Alarm occurred
M31	Continuous residual leakage current overlimit	BOOL	0: No alarm; 1: Alarm occurred
M32	Local emergency stop fault	BOOL	0: No alarm; 1: Alarm occurred

3.12 Alarm Status 4 (warningGGIO4)

Param.	Fault Name	Data Type	Description
N01	Remote emergency stop fault	BOOL	0: No alarm; 1: Alarm occurred
N02	BMS emergency stop fault	BOOL	0: No alarm; 1: Alarm occurred
N03	Phase-to-PE fault	BOOL	0: No alarm; 1: Alarm occurred
N04	Capacitor voltage resonance fault	BOOL	0: No alarm; 1: Alarm occurred
N05	High humidity	BOOL	0: No alarm; 1: Alarm occurred
N06	Humidity detection disconnected	BOOL	0: No alarm; 1: Alarm occurred
N07	Temperature too high	BOOL	0: No alarm; 1: Alarm occurred
N08	Temperature detection disconnected	BOOL	0: No alarm; 1: Alarm occurred
N09	Fuse fault	BOOL	0: No alarm; 1: Alarm occurred
N10	Cabinet door fault	BOOL	0: No alarm; 1: Alarm occurred
N11	Water pump locked-rotor	BOOL	0: No alarm; 1: Alarm occurred
N12	Water pump speed regulation signal absent	BOOL	0: No alarm; 1: Alarm occurred
N13	Water pump idling	BOOL	0: No alarm; 1: Alarm occurred
N14	Water cooling system hydraulic fault	BOOL	0: No alarm; 1: Alarm occurred
N15	Fan power sequence incorrect	BOOL	0: No alarm; 1: Alarm occurred

N16	Fan power disconnected	BOOL	0: No alarm; 1: Alarm occurred
N17	Three-phase fan fault	BOOL	0: No alarm; 1: Alarm occurred
N18	Heater contactor fault	BOOL	0: No alarm; 1: Alarm occurred
N19	Water pump power disconnected	BOOL	0: No alarm; 1: Alarm occurred
N20	Software version inconsistent between master and slave	BOOL	0: No alarm; 1: Alarm occurred
N21	Dynamic environment fault	BOOL	0: No alarm; 1: Alarm occurred
N22	DC side midpoint sampling disconnected	BOOL	0: No alarm; 1: Alarm occurred
N23	DC side voltage sampling disconnected	BOOL	0: No alarm; 1: Alarm occurred
N24	AC and DC voltage mismatch	BOOL	0: No alarm; 1: Alarm occurred
N25	Grid voltage mismatch	BOOL	0: No alarm; 1: Alarm occurred
N26	Heat exchanger fan fault	BOOL	0: No alarm; 1: Alarm occurred
N27	DC circuit breaker fan fault	BOOL	0: No alarm; 1: Alarm occurred
N28	AC circuit breaker fan fault	BOOL	0: No alarm; 1: Alarm occurred
N29	Master and slave identification fault	BOOL	0: No alarm; 1: Alarm occurred
N30	Power inconsistent between master and slave	BOOL	0: No alarm; 1: Alarm occurred
N31	IGBT temperature low	BOOL	0: No alarm; 1: Alarm occurred
N32	Zero sequence loop current too large	BOOL	0: No alarm; 1: Alarm occurred

3.13 Alarm Status 5 (warningGGIO5)

Param.	Fault Name	Data Type	Description
O01	FPGA side fault	BOOL	0: No alarm; 1: Alarm occurred
O02	Fault of parallel devices in cabinet	BOOL	0: No alarm; 1: Alarm occurred
O03	Fiber communication interrupted	BOOL	0: No alarm; 1: Alarm occurred
O04	Fiber communication CRC fault	BOOL	0: No alarm; 1: Alarm occurred
O05	Control mode inconsistent	BOOL	0: No alarm; 1: Alarm occurred
O06	Capacitor harmonic overlimit	BOOL	0: No alarm; 1: Alarm occurred
O07	Fan pump and water pump switch error	BOOL	0: No alarm; 1: Alarm occurred
O08	PCS interaction fault	BOOL	0: No alarm; 1: Alarm occurred
O09	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O10	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O11	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O12	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O13	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O14	Reserved	BOOL	0: No alarm; 1: Alarm occurred

O15	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O16	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O17	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O18	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O19	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O20	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O21	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O22	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O23	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O24	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O25	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O26	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O27	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O28	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O29	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O30	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O31	Reserved	BOOL	0: No alarm; 1: Alarm occurred
O32	Reserved	BOOL	0: No alarm; 1: Alarm occurred

4 GOOSE Data (PIGO)

4.1 Remote Measurement Parameters (measGGIO1)

Param.	Name	Data Type	Description	Lower Limit	Upper Limit	Default	Unit
P01	Rated power	Float32	-	0	65535	1725	1 kW
P02	Battery pack SOC	Float32	-	0	100.0	0	0.1%
P03	PCS system status–value mode	Int32	0: Stopped 1: Standby 2: On-grid charging operation 3: On-grid discharging operation 4: On-grid zero power operation 5: Off-grid operation 6: Fault status	0	65535	0	1
P04	Total active output power	Float32	-	–32768	32767	0	1 kW
P05	Total reactive output power	Float32	-	–32768	32767	0	1 kVar
P06	Maximum charging power of battery pack	Float32	-	0	65535	0	1 kW
P07	Maximum discharging power of battery pack	Float32	-	0	65535	0	1 kW

P08	Active power setting	Float32	Returns actual value; used to determine if setting is successful.	–32768	32767	0	1 kW
P09	Reactive power setting	Float32	Returns actual value; used to determine if setting is successful.	–32768	32767	0	1 kVar

Partner EGS Headquarter
www.partneregs.com/www.wattox.com

Add.: İnönü Mahallesi 1748. Cadde No:1 Batıkent,
6370 Yenimahalle, Ankara, Türkiye
Tel: +90 312 920 00 18

Partner EGS Factory
www.partneregs.com/www.wattox.com

Add.: Alıcı Mah. 2007 Cad. No:7
06909 Sincan / Ankara, Türkiye
Tel: +90 312 256 00 86
Fax: info@wattox.com